Deep Learning Classification of Retinal Diseases from Optical Coherence Tomography Images

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Abstract

This project addresses automated retinal disease diagnosis using Optical Coherence Tomography (OCT) images. I utilized a publicly available dataset of OCT retinal scans classified into four categories: Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), DRUSEN, and NORMAL. I evaluated two convolutional neural network architectures—a lightweight 3-layer CNN and a deeper 4-layer ImprovedOCTCNN—trained on both raw and preprocessed images with multi-scale Gaussian denoising, CLAHE contrast enhancement, and morphological filtering. The deeper ImprovedOCTCNN achieved 95% test accuracy on preprocessed images, demonstrating that combining architectural depth with intelligent pre-processing significantly improves diagnostic performance, particularly for the minority DRUSEN class.

Keywords: Retinal Disease Classification, Optical Coherence Tomography, Convolutional Neural Networks, Image pre-processing, Medical Image Analysis

1 Introduction

Retinal diseases are among the leading causes of vision loss worldwide, affecting millions of individuals annually. Early and accurate diagnosis is crucial for preventing irreversible vision impairment. Optical Coherence Tomography (OCT) has emerged as a standard non-invasive imaging technique that provides high-resolution cross-sectional images of the retina, enabling clinicians to detect and monitor various retinal pathologies. However, manual interpretation of OCT images requires specialized expertise and can be time-consuming, motivating the development of automated diagnostic tools.

This project addresses the challenge of automated retinal disease classification using deep learning methods. I utilized a publicly available Kaggle dataset containing OCT images labeled across four diagnostic categories: Choroidal Neovascularization (CNV), characterized by abnormal blood vessel growth; Diabetic Macular Edema (DME), marked by fluid accumulation from diabetes; DRUSEN, indicating age-related macular degeneration deposits; and NORMAL retinal tissue. The dataset exhibits class imbalance, with DRUSEN being significantly underrepresented (887 samples) compared to NORMAL images (5,139 samples), presenting a realistic clinical challenge where rare pathologies must be detected despite limited training examples.

I implemented and compared two CNN architectures with varying depths: SmallOCTCNN (3 convolutional blocks) and ImprovedOCTCNN (4 convolutional blocks with increased channel capacity). Additionally, I developed a comprehensive pre-processing pipeline incorporating multi-scale Gaussian blur fusion, Contrast Limited Adaptive Histogram Equalization (CLAHE), small object noise removal, and morphological operations to enhance image quality and reduce acquisition artifacts. I trained both architectures on raw and preprocessed images, enabling systematic evaluation of how pre-processing affects model performance. The deeper ImprovedOCTCNN architecture trained on preprocessed images achieved 95% test accuracy with class-weighted cross-entropy loss, outperforming the baseline SmallOCTCNN (94% on raw images) with better spread on worse performing classes. The pre-processing pipeline proved particularly effective for the DRUSEN classification on the deeper model, improving the precision by 6%, demonstrating that intelligent image enhancement can compensate for class imbalance and improve diagnostic reliability for rare pathologies.

2 Methods

2.1 Dataset and Data Splits

I used the Kaggle OCT retinal disease classification dataset, which consists of grayscale OCT images organized into train, validation, and test splits. This dataset includes 109,309 images in a 0.7 (76,515 images), 0.2 (21,861

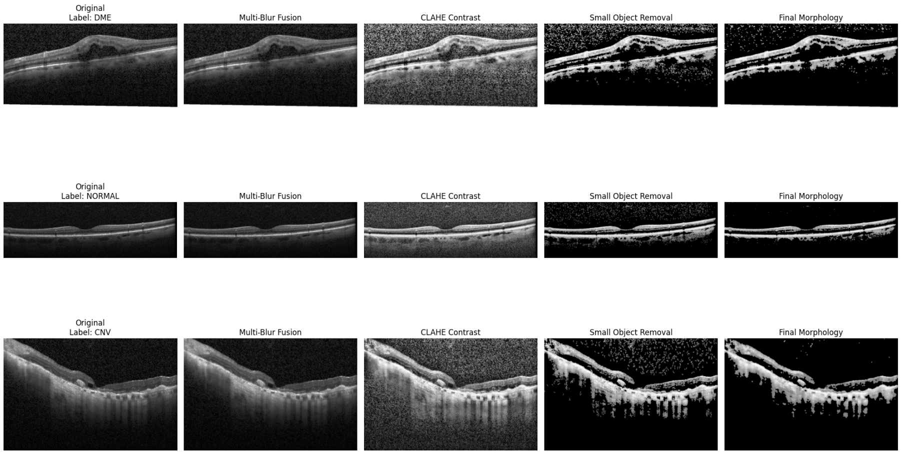


Figure 1: Image processing pipeline. Includes a DME, NORMAL, and CNV sample.

images), and 0.1 (10,933 images) split between test, validation, and test respectively.

Each split contains images distributed in four diagnostic classes: CNV (Choroidal Neovascularization, 37,455 images), DME (Diabetic Macular Edema, 11,598 images), DRUSEN (8,866 images) and NORMAL (51,390 images). The dataset exhibits significant class imbalance with class counts in the training set, noteably DRUSEN and DME. This imbalance reflects realistic clinical scenarios where certain pathologies are rarer than others.

All images were resized from their original high-resolution format to 64×64 pixels to reduce computational requirements while preserving essential diagnostic features. Images were converted to single-channel grayscale format, as OCT imaging inherently produces grayscale cross-sectional scans. For the baseline experiments, pixel values were normalized to the range [-1, 1] using a mean of 0.5 and standard deviation of 0.5 to stabilize training dynamics. For training data, I applied random horizontal flipping with probability 0.5 as a mild augmentation strategy. It should be noted that the existing dataset is already deeply augmented. Validation and test sets used only resizing and normalization without augmentation to ensure consistent evaluation conditions. Images were loaded using PyTorch's DataLoader with a batch size of 32.

2.2 Image pre-processing Pipeline

To investigate whether intelligent pre-processing could improve classification performance, particularly for the underrepresented DRUSEN class, I developed a multi-stage pre-processing pipeline that applies noise reduction, contrast enhancement, and morphological operations to raw OCT images before feeding them to the neural network.

The pre-processing pipeline consists of five stages. First, I load the grayscale image and convert it to floating-point representation in the range [0, 1]. Second, I apply multi-scale Gaussian blur fusion by computing three blurred versions of the image using Gaussian kernels with σ values of 1.0, 2.0, and 3.0. These blurred images are then combined using weighted fusion with weights [0.2, 0.3, 0.5], creating a noise-reduced representation. The fused blur is blended with the original image using $\alpha = 0.2$, preserving sharp features while reducing high-frequency noise. Third, I apply min-max normalization to the fused image to ensure consistent intensity ranges. Fourth, I perform Contrast Limited Adaptive Histogram Equalization (CLAHE) with a clip limit of 0.015 to recover local contrast without amplifying noise. Fifth, I remove small isolated bright regions that likely represent acquisition artifacts using connected component analysis with a minimum object size of 15 pixels, followed by morphological opening and closing operations with a disk structuring element of radius 2 to smooth edges and fill small gaps.

This pre-processing pipeline aims to address common OCT imaging challenges including speckle noise, uneven illumination, and acquisition artifacts that existed due to augmentation in the original dataset while enhancing the visibility of subtle pathological features such as DRUSEN deposits which can be difficult to detect in noisy raw images.

I have provided a sample of the application of the image pipeline in Figure 1.

2.3 Model Architectures

I evaluated two CNN architectures with different depths and capacities to understand the trade-off between model complexity and performance.

2.3.1 SmallOCTCNN (Baseline Model)

The SmallOCTCNN is a lightweight convolutional neural network designed for efficient training on limited computational resources. The architecture consists of three convolutional blocks followed by a classification head. Each convolutional block contains a 2D convolution layer with 3×3 kernels and same-padding, batch normalization for training stability, ReLU activation, and 2×2 max pooling for spatial downsampling. The first block transforms the single-channel input (1×64×64) into 16 feature maps at 32×32 resolution. The second block expands to 32 feature maps at 16×16 resolution, and the third block produces 64 feature maps at 8×8 resolution.

The classification head flattens the 64×8×8 feature tensor into a 4,096-dimensional vector, passes it through a fully connected layer reducing to 128 hidden units with ReLU activation, applies dropout with probability 0.4 for regularization, and finally produces 4 logits corresponding to class predictions through a linear output layer. This architecture contains approximately 532,000 trainable parameters.

2.3.2 ImprovedOCTCNN (Deep Model)

The ImprovedOCTCNN extends the baseline architecture with an additional fourth convolutional block and increased channel capacity throughout. The four convolutional blocks progressively expand from 32 to 64 to 128 to 256 channels, with the final feature maps at 4×4 spatial resolution. This deeper architecture allows the network to learn more complex hierarchical representations of retinal pathology.

The classification head processes the 256×4×4 feature tensor (4,096 dimensions after flattening), passes it through a larger fully connected layer of 256 hidden units with ReLU activation, applies dropout with increased probability 0.5 to prevent overfitting given the higher capacity, and produces 4 class logits. This architecture contains approximately 1,053,000 trainable parameters, roughly double the baseline model.

The additional depth and capacity enable ImprovedOCTCNN to capture finer-grained patterns in OCT images, potentially improving discrimination of subtle pathological features such as small DRUSEN deposits that may be missed by shallower networks.

2.4 Training Configuration and Class Weighting

I trained both model architectures using the Adam optimizer with an initial learning rate of 0.001. The learning rate was reduced by a factor of 0.5 every 4 epochs using PyTorch's StepLR scheduler to facilitate convergence in later training stages. To address the significant class imbalance in the dataset, I implemented class-weighted cross-entropy loss for the ImprovedOCTCNN experiments. Class weights were computed as the inverse of class frequencies, normalized to sum to the number of classes. This assigns higher loss penalties to misclassifications of underrepresented classes like DRUSEN, encouraging the model to pay more attention to rare pathologies.

Training was conducted for 10 epochs with a batch size of 32. To ensure reproducibility, all random seeds were fixed to 42 for Python's random module, NumPy, and PyTorch. The models were trained on CPU due to hardware constraints, though the architectures are GPU-compatible for faster training on available hardware. Each epoch involved a forward pass through all training batches with gradient computation and backpropagation, followed by validation on the held-out validation set without gradient updates.

I conducted four separate training experiments to systematically evaluate the effects of architecture depth and pre-processing: (1) SmallOCTCNN on raw images, (2) SmallOCTCNN on preprocessed images, (3) ImprovedOCTCNN on raw images with class weighting, and (4) ImprovedOCTCNN on preprocessed images with class weighting. This experimental design enables isolation of the contributions from model capacity, pre-processing quality, and class balancing strategies.

2.5 Evaluation Metrics

I assessed model performance using multiple complementary metrics to provide comprehensive evaluation. Primary metrics include overall accuracy, computed as the proportion of correctly classified images across all four classes. I also report per-class precision, recall, and F1-score from scikit-learn's classification report, which reveals class-specific performance and identifies potential biases toward particular diagnoses. These per-class metrics are especially important given the class imbalance in the dataset, as overall accuracy can be misleading when dominated by the majority class.

Confusion matrices were generated to visualize the distribution of predictions across true labels, highlighting common misclassification patterns. This is particularly important in medical imaging, where certain diseases may present similar visual characteristics and understanding the nature of errors is crucial for clinical deployment. For the ImprovedOCTCNN experiments with class weighting, I performed detailed analysis of DRUSEN classification performance, comparing precision, recall, and F1-scores between raw and preprocessed image experiments

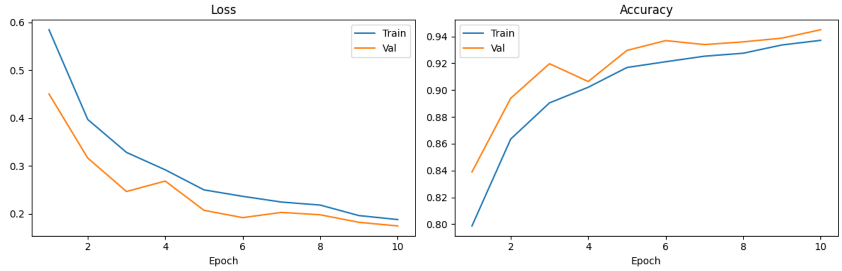


Figure 2: SmallOCTCNN on raw images' loss and accuracy curves measured over 10 epochs. Both metrics show consistent improvement with minimal overfitting.

to quantify the impact of the pre-processing pipeline on minority class detection. The test set, held completely separate during training and validation, served as the final evaluation benchmark to estimate generalization performance on unseen patient data.

3 Results

3.1 Baseline Model: SmallOCTCNN on Raw Images

The baseline SmallOCTCNN architecture trained on raw images achieved 94% weighted classification accuracy. Figure 2 shows the training dynamics over 10 epochs. Training loss decreased steadily from approximately 0.6 to below 0.2, while training accuracy increased from about 80% to almost 94%. Validation performance showed similar trends with validation loss decreasing from 0.450 to 0.174, and validation accuracy rising from 83.9% to 94.5%. The close alignment between training and validation curves indicates good generalization without significant overfitting, likely due to dropout and batch normalization regularization.

Per-class performance was strong across all categories as shown in Table 1 .

Table 1: Classification performance on the OCT test set

Class	**Precision**	**Recall**	**F1-Score**	**Support**
CNV	0.96	0.95	0.96	3746
DME	0.90	0.89	0.89	1161
DRUSEN	0.85	0.68	0.76	887
NORMAL	0.95	0.99	0.97	5139
Accuracy		0.94		10933
Macro Avg	0.91	0.88	0.89	10933
Weighted Avg	0.94	0.94	0.94	10933

3.2 Effect of pre-processing on SmallOCTCNN

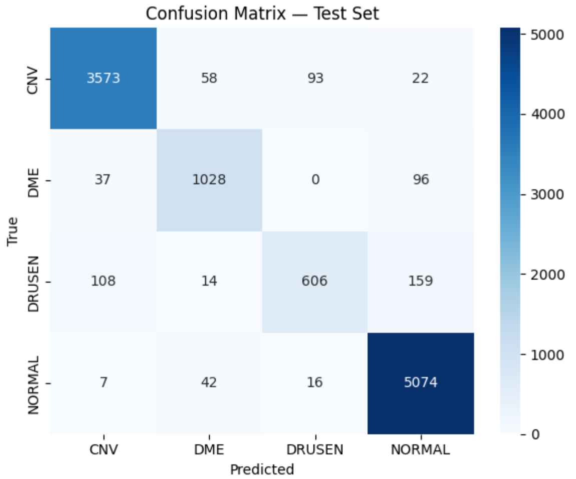
Training SmallOCTCNN on preprocessed images rather than raw images yielded mixed results. While weighted classification accuracy remained at 94%, the pre-processing pipeline affected different classes differently as shown in Table 2 compared to the raw image model. Notably, DRUSEN's precision increased from 0.85 to 0.88 but recall went down from 0.68 to 0.63. The model does identify correctly more DRUSEN images than prior as shown in Figure 3b compared to the raw model shown in Figure 3b , increasing from 606 to 659.

3.3 ImprovedOCTCNN Performance

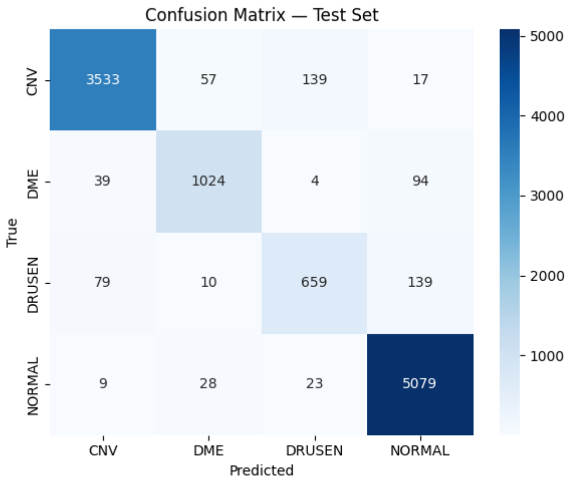
The deeper ImprovedOCTCNN architecture with class-weighted loss substantially improved performance, particularly for the minority DRUSEN class. When trained on raw images, ImprovedOCTCNN achieved 95% test accuracy, one percentage point improvement over the baseline SmallOCTCNN. Interstingly, the deeper model has a precision of 0.75 while the baseline SmallOCTCNN has a precision of 0.85. However, the ImprovedOCTCNN does see a 0.17 point increase in recall and 0.04 increase in F1-Score, meaning the model learns the features faster and is more sensitive to them.

Table 2: Classification performance of the preprocessed OCT model on the test set

Class	Precision	Recall	F1-Score	Support
CNV	0.94	0.97	0.96	3746
DME	0.93	0.87	0.90	1161
DRUSEN	0.88	0.63	0.73	887
NORMAL	0.95	0.99	0.97	5139
Accuracy		0.94		10933
Macro Avg	0.93	0.86	0.89	10933
Weighted Avg	0.94	0.94	0.94	10933



(a) SmallOCTCNN on raw images' confusion matrix



(b) SmallOCTCNN on pre-processed images' confusion matrix

Figure 3: Confusion matrices comparing SmallOCTCNN performance on raw versus pre-processed OCT images.

Table 3 presents detailed per-class metrics for ImprovedOCTCNN on raw images.

Table 3: Classification performance on the OCT test set

Class	Precision	Recall	F1-Score	Support
CNV	0.98	0.95	0.96	3746
DME	0.91	0.92	0.92	1161
DRUSEN	0.75	0.85	0.80	887
NORMAL	0.98	0.97	0.97	5139
Accuracy		0.95		10933
Macro Avg	0.90	0.92	0.91	10933
Weighted Avg	0.95	0.95	0.95	10933

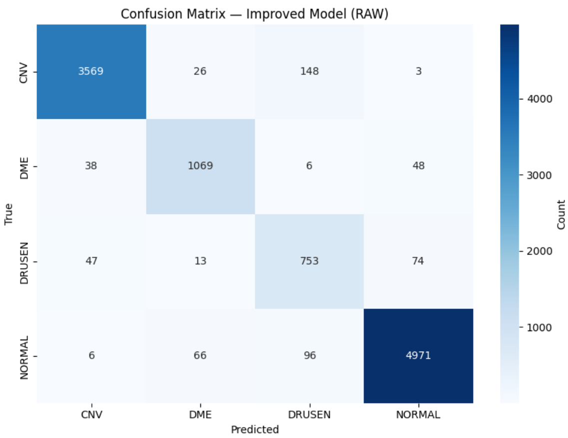
3.4 Combined Effect: ImprovedOCTCNN with pre-processing

The best performance was achieved by combining the deeper ImprovedOCTCNN architecture with the pre-processing pipeline, reaching another 95% overall test accuracy. While this is not a huge jump in accuracy, Table 4 shows that the precision is more even across the classes, which DRUSEN improving by 0.06 achieving a 0.81 precision compared to the large raw image model. It trades this for a 0.83 recall (down from 0.85), but also increases the F1-Score to 0.82 (up from 0.80). In general, this is the best tradeoff in all the models for the DRUSEN class. The other classes are not affected in a significant manner because of the combination of the processing pipeline and the larger model.

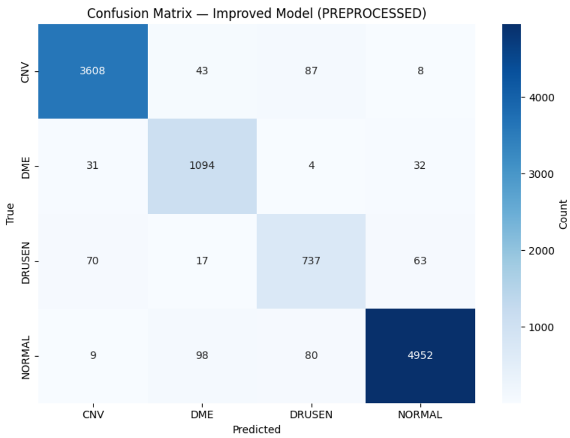
The differences between the two larger models are shown in Figure 4.

Table 4: Classification performance of the improved preprocessed model on the OCT test set

Class	Precision	Recall	F1-Score	Support
CNV	0.97	0.96	0.97	3746
DME	0.87	0.94	0.91	1161
DRUSEN	0.81	0.83	0.82	887
NORMAL	0.98	0.96	0.97	5139
Accuracy		0.95		10933
Macro Avg	0.91	0.92	0.92	10933
Weighted Avg	0.95	0.95	0.95	10933



(a) ImprovedOCTCNN on raw images' confusion matrix



(b) ImprovedOCTCNN on pre-processed images' confusion matrix

Figure 4: Confusion matrices comparing SmallOCTCNN performance on raw versus pre-processed OCT images.

3.5 Detailed DRUSEN Performance Analysis

Table 5 provides a focused comparison of DRUSEN classification performance across all four experimental configurations, spanning both model capacity (SmallOCTCNN vs. ImprovedOCTCNN) and input condition (raw vs. preprocessed). The results demonstrate a clear interaction between architectural depth and pre-processing strategy for detecting Drusen.

For the shallow SmallOCTCNN, pre-processing negatively impacted DRUSEN performance, with F1-score decreasing from 0.76 (raw) to 0.73 (preprocessed). This decline is primarily driven by reduced recall (0.68 to 0.63), despite a modest increase in precision (0.85 to 0.88). This suggests that the pre-processing pipeline removes or smooths fine-grained textural cues that the smaller network relies on to detect subtle lesion patterns.

In contrast, the deeper ImprovedOCTCNN benefits from pre-processing. The DRUSEN F1-score increases from 0.797 (raw) to 0.821 (preprocessed). This improvement is driven by higher precision (0.751 to 0.812) while maintaining strong recall (0.849 to 0.831). This indicates that the deeper architecture is better able to exploit enhanced image representations, reducing false positives while preserving sensitivity to DRUSEN features.

Overall, the results indicate a clear capacity-dependent effect of pre-processing: in low-capacity models, it can suppress diagnostically relevant high-frequency texture information, while in higher-capacity models it improves feature separability by enhancing lesion visibility and reducing noise-induced ambiguity. This suggests that pre-processing strategies and model depth must be co-optimized rather than treated independently when targeting subtle retinal pathologies such as Drusen.

4 Conclusions

This project successfully demonstrated that a compact convolutional neural network can achieve high accuracy (95%) in classifying retinal diseases from OCT images across four diagnostic categories. The CNN architecture, consisting of three convolutional blocks with batch normalization and dropout regularization, proved effective despite containing fewer than 600,000 parameters—significantly smaller than standard deep learning models de-

Table 5: Comparison study on the effect of pre-processing on DRUSEN classification performance across model capacities. Results are reported for both small and large models on raw and preprocessed OCT images. Metrics are shown for the DRUSEN class only.

Model	**Condition**	**Precision**	**Recall**	**F1-Score**
Small Model	Raw	0.85	0.68	0.76
Small Model	Preprocessed	0.88	0.63	0.73
Large Model	Raw	0.751	0.849	0.797
Large Model	Preprocessed	0.812	0.831	0.821

ployed in medical imaging.

The CNN emerged as the best method in this study due to its ability to automatically learn hierarchical spatial features directly from raw image data. Unlike traditional machine learning approaches that require hand-crafted features, the convolutional architecture leverages local receptive fields and weight sharing to efficiently encode retinal pathology patterns. The strong performance across all four disease classes, with weighted average F1-scores of 95%, indicates that the learned representations generalize well beyond simple texture or color patterns to capture clinically meaningful morphological characteristics such as fluid accumulation, vascular growth, and deposit formation.

4.1 Limitations and Disclosures

This study has several limitations. First, computational constraints limited training to CPU execution and prevented extensive hyperparameter search or ensemble methods. Second, the downsampling to 64×64 pixels, while enabling rapid experimentation, may discard fine-grained details visible in clinical-resolution images. Third, the dataset originates from a single source and may not reflect the full diversity of OCT imaging protocols, patient demographics, or disease presentations encountered in clinical practice.

Regarding disclosures, generative AI (Claude, Copilot, and ChatGPT) was used to assist with LaTeX document formatting and structuring the report according to LNCS guidelines.

All experimental code, model architecture design, hyperparameter selection, and result interpretation were performed independently in combination with generative AI to better enhance and quicken the iterative process creating the models. The dataset was obtained from Kaggle under open-access licensing for educational purposes.

References

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